

Run report: every case, every experiment

Every held-out case, every experiment, from run 2026-10-08-heldout. Generated from the recorded rows; nothing typed by hand.

S2

S2 · Production AI Engineering

9 min read · Published 2026-10-08 · Run report

This report is generated by `tools/build_report_source.py` from `enterprise_knowledge_rag_poc/runs/2026-10-08-heldout/`. Labels are in `groundtruth/labels.yaml`; definitions in the technical edition §4.1 and §7.

A • Retrieval: needed index units in the top 10 (no filter)

case	needed	vector	BM25	hybrid	invalid in vector top 5
H-K0	2	2/2	2/2	2/2	1
H-K1	2	2/2	2/2	2/2	3
H-K2	2	2/2	2/2	2/2	1
H-K3	2	2/2	2/2	2/2	2
H-K4	4	4/4	3/4	4/4	2
H-K5	4	0/4	1/4	1/4	3
H-K6	2	1/2	1/2	1/2	2
H-K7	2	2/2	2/2	2/2	1
H-K8	1	0/1	0/1	0/1	4
H-K9	2	0/2	0/2	0/2	0
H-K10	3	3/3	2/3	2/3	3
H-K11	1	1/1	1/1	1/1	1
H-K12	1	1/1	1/1	1/1	1
H-K13	2	0/2	2/2	1/2	3
H-K15	1	0/1	0/1	0/1	1
H-P1	1	1/1	1/1	1/1	0
H-P2	3	3/3	2/3	2/3	2
H-P3	1	1/1	1/1	1/1	2

A2 • structured records. For each needed record: returned by the structured route; exact fields found together in a top-20 prose unit.

case	record	route	vector	BM25	hybrid
H-K0	dep:acme/production/search-api/products-2026.09.22-1	✓	✗	✗	✗
H-K5	dep:acme/production/checkout-api/4.17.0	✓	✗	✗	✗
H-K7	cmdb:acme/production/notification-service	✓	✗	✗	✗
H-K8	dep:acme/production/inventory-api/3.9.0	✓	✗	✗	✗
H-K9	dep:acme/production/checkout-api/4.17.0	✓	✗	✗	✗
H-K9	cmdb:acme/production/checkout-api	✓	✗	✗	✗
H-K12	cmdb:acme/production/payment-gateway	✓	✗	✗	✗
H-K14b	cmdb:acme/production/recommendation-service	✓	✗	✗	✗
H-P2	dep:globex/production/checkout-api/7.4.0	✓	✗	✗	✗

B · Admission over the same frozen candidates

case	pool invalid	naive: invalid in context	naive: unreadable	governed: invalid	governed: unreadable	needed kept naive	needed kept governed
H-K0	1	1	0	0	0	2/2	2/2
H-K1	10	4	1	0	0	2/2	2/2
H-K2	2	1	1	0	0	2/2	2/2
H-K3	4	3	0	0	0	2/2	2/2
H-K4	7	3	0	0	0	4/4	4/4
H-K5	8	4	1	0	0	0/3	3/3
H-K6	8	3	0	0	0	1/2	2/2
H-K7	1	1	1	0	0	2/2	2/2
H-K8	7	4	0	0	0	0/0	0/0
H-K9	7	4	1	0	0	0/0	0/0
H-K10	8	3	0	0	0	2/3	3/3
H-K11	2	2	2	0	0	1/1	1/1
H-K12	3	3	1	0	0	1/1	1/1
H-K13	9	5	0	0	0	1/2	2/2
H-K14	1	0	2	0	0	0/0	0/0
H-K15	3	1	2	0	0	0/1	1/1
H-K12b	1	1	1	0	0	0/0	0/0
H-K12c	11	5	0	0	0	0/0	0/0
H-K14b	6	2	1	0	0	0/0	0/0
H-P1	0	0	0	0	0	1/1	1/1
H-P2	10	5	0	0	0	2/3	3/3
H-P3	1	1	1	0	0	1/1	1/1

Invalid units the naive arm let in, by case:

- **H-K0:** GX-RB-SRCH-001@v1#recovery
- **H-K1:** GX-RB-CHK-003@v2#remediation, RB-CART-003@v2#remediation, RB-CHK-007@v3#remediation, RB-CHK-007@v5#remediation
- **H-K2:** STG-OPS-004@v3#scaling
- **H-K3:** GX-RB-INV-002@v1#remediation, RB-INV-004@v4#remediation, RB-PRC-002@v1#remediation
- **H-K4:** RB-CHK-007@v3#remediation, RB-CHK-007@v5#remediation, RB-CHKW-002@v1#remediation
- **H-K5:** GUIDE-PERF-031@v1#pool-saturation, RB-CHK-007@v3#remediation, RB-CHK-007@v3#symptoms, STG-OPS-004@v3#pool-and-connection-settings
- **H-K6:** RB-CHK-007@v3#remediation, WIKI-CHK-FAQ@v1#what-fixes-checkout-latency-fastest, WIKI-CHK-TIPS@v1#latency-after-a-deploy
- **H-K7:** RB-NOT-004@v1#remediation

- **H-K8:** RB-CHK-007@v3#remediation , RB-CHK-007@v4#approval-and-exceptions , RB-CHK-007@v4#remediation , RB-CHK-007@v5#remediation
- **H-K9:** INC-4911@v1#summary , RB-CHK-007@v3#remediation , RB-CHK-007@v3#symptoms , RB-CHK-007@v5#status
- **H-K10:** RB-CHK-007@v3#remediation , RB-CHK-007@v3#symptoms , RB-CHK-007@v5#status
- **H-K11:** SEC-PM-2207@v1#remediation , SEC-PM-2207@v1#root-cause
- **H-K12:** RB-PAY-002@v3#recovery , RB-PAY-002@v3#remediation , VENDOR-ACN-01@v1#escalation
- **H-K13:** GX-RB-CHK-003@v2#remediation , RB-CHK-007@v3#remediation , RB-CHK-007@v3#symptoms , RB-CHK-007@v5#status , WIKI-CHK-FAQ@v1#what-fixes-checkout-latency-fastest
- **H-K15:** GUIDE-PERF-031@v1#pool-saturation
- **H-K12b:** VENDOR-ACN-01@v1#escalation
- **H-K12c:** RB-CART-003@v2#diagnose , RB-CART-003@v2#remediation , RB-CHK-007@v3#symptoms , RB-CHK-007@v4#approval-and-exceptions , RB-CHK-007@v4#symptoms
- **H-K14b:** INC-4702@v1#summary , RB-INV-004@v5#cache-caution
- **H-P2:** RB-CHK-007@v3#remediation , RB-CHK-007@v3#symptoms , RB-CHK-007@v4#diagnose , RB-CHK-007@v4#symptoms , RB-CHK-007@v5#remediation
- **H-P3:** GUIDE-PERF-031@v1#pool-saturation

C · Packing, same admitted pool: needed units fully present · qualifiers kept

case	300 truncate	300 assembler	450 truncate	450 assembler	600 truncate	600 assembler	900 truncate	900 assembler
H-K0	2/3 · q 1/1	3/3 · q 1/1	2/3 · q 1/1	3/3 · q 1/1	2/3 · q 1/1	3/3 · q 1/1	3/3 · q 1/1	3/3 · q 1/1
H-K1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1
H-K2	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1
H-K3	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1
H-K4	0/4 · q 1/1	1/4 · q 1/1	2/4 · q 1/1	1/4 · q 1/1	4/4 · q 1/1	2/4 · q 1/1	4/4 · q 1/1	3/4 · q 1/1
H-K5	0/5 · q 0/1	1/5 · q 0/1	1/5 · q 1/1	2/5 · q 0/1	2/5 · q 1/1	3/5 · q 0/1	4/5 · q 1/1	4/5 · q 1/1
H-K6	1/2 · q 1/1	2/2 · q 1/1	1/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1	2/2 · q 1/1
H-K7	1/3 · q 0/1	1/3 · q 0/1	2/3 · q 1/1	1/3 · q 0/1	2/3 · q 1/1	3/3 · q 1/1	3/3 · q 1/1	3/3 · q 1/1
H-K8	0/1 · q 0/2	1/1 · q 1/2	0/1 · q 0/2	1/1 · q 1/2	1/1 · q 1/2	1/1 · q 1/2	1/1 · q 1/2	1/1 · q 1/2
H-K9	2/4 · q 1/1	2/4 · q 1/1	2/4 · q 1/1	4/4 · q 1/1	2/4 · q 1/1	4/4 · q 1/1	2/4 · q 1/1	4/4 · q 1/1
H-K10	2/3	1/3	2/3	1/3	2/3	2/3	3/3	2/3
H-K11	1/1	0/1	1/1	1/1	1/1	1/1	1/1	1/1
H-K12	1/2	1/2	1/2	1/2	1/2	1/2	1/2	1/2
H-K13	1/2	1/2	2/2	1/2	2/2	1/2	2/2	2/2
H-K14	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0
H-K15	0/1	0/1	0/1	0/1	0/1	0/1	1/1	1/1
H-K12b	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0
H-K12c	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
H-K14b	0/1	1/1	0/1	1/1	0/1	1/1	1/1	1/1
H-P1	1/1	0/1	1/1	0/1	1/1	0/1	1/1	1/1
H-P2	3/4	3/4	3/4	4/4	4/4	4/4	4/4	4/4
H-P3	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1

D · End to end, 3 seed(s): correct runs (naive · governed raw · governed bound)

case	naive	governed raw	governed bound	naive leaks	naive forbidden action	governed binding changed
H-K0	3/3	3/3	3/3	0	0	1
H-K1	0/3	0/3	0/3	0	0	3
H-K2	3/3	3/3	3/3	0	0	0
H-K3	3/3	3/3	3/3	0	0	0
H-K4	2/3	3/3	3/3	0	0	3
H-K5	0/3	0/3	0/3	0	3	3
H-K6	0/3	3/3	3/3	0	0	2
H-K7	3/3	3/3	3/3	0	0	3
H-K8	0/3	0/3	0/3	0	0	3
H-K9	0/3	0/3	0/3	0	2	3
H-K10	0/3	1/3	1/3	0	3	1
H-K11	0/3	3/3	3/3	3	0	0
H-K12	2/3	3/3	3/3	0	0	1
H-K13	0/3	0/3	0/3	0	3	3
H-K14	3/3	3/3	3/3	0	0	0
H-K15	0/3	1/3	1/3	0	3	0
H-K12b	0/3	3/3	3/3	3	0	0
H-K12c	0/3	3/3	3/3	0	0	0
H-K14b	3/3	3/3	3/3	0	0	0
H-P1	3/3	0/3	0/3	0	0	0
H-P2	3/3	3/3	3/3	0	0	0
H-P3	3/3	3/3	3/3	0	0	0

Every answer, first seed (status · action · approval):

case	naive	governed bound
H-K0	✓ answer · rollback_index_alias · approval	✓ answer · rollback_index_alias · approval
H-K1	✗ answer · enable_flag	✗ partial · none
H-K2	✓ answer · scale_out	✓ answer · scale_out
H-K3	✓ answer · resync_job	✓ answer · resync_job
H-K4	✗ partial · rollback_release · approval	✓ answer · rollback_release · approval
H-K5	✗ answer · raise_pool_size	✗ partial · none · approval
H-K6	✗ escalate · escalate	✓ answer · rollback_release · approval
H-K7	✓ escalate · escalate · approval	✓ escalate · escalate · approval
H-K8	✗ abstain · none	✗ abstain · none
H-K9	✗ partial · restart_pods · approval	✗ answer · rollback_release · approval
H-K10	✗ answer · restart_pods	✓ partial · rollback_release · approval
H-K11	✗ answer · none	✓ abstain · none
H-K12	✓ escalate · escalate	✓ escalate · escalate
H-K13	✗ answer · restart_pods	✗ partial · none
H-K14	✓ abstain · none	✓ abstain · none
H-K15	✗ answer · raise_pool_size	✓ answer · none
H-K12b	✗ answer · none	✓ abstain · none
H-K12c	✗ answer · rollback_release · approval	✓ answer · enable_flag
H-K14b	✓ abstain · none	✓ abstain · none
H-P1	✓ answer · none	✗ abstain · none
H-P2	✓ answer · switch_traffic	✓ answer · switch_traffic
H-P3	✓ answer · raise_pool_size	✓ answer · raise_pool_size

D • Sensitivity model (one seed): correct

case	naive	governed raw	governed bound
H-K0	✗	✓	✓
H-K1	✓	✓	✓
H-K2	✓	✓	✓
H-K3	✓	✓	✓
H-K4	✓	✓	✓
H-K5	✗	✗	✗
H-K6	✗	✓	✓
H-K7	✗	✓	✓
H-K8	✗	✗	✗
H-K9	✗	✗	✗
H-K10	✗	✓	✓
H-K11	✗	✓	✓
H-K12	✓	✓	✓
H-K13	✗	✗	✗
H-K14	✓	✓	✓
H-K15	✗	✓	✓
H-K12b	✗	✓	✓
H-K12c	✗	✓	✓
H-K14b	✓	✓	✓
H-P1	✓	✗	✗
H-P2	✓	✓	✓
H-P3	✓	✓	✓

D2 · Verifier and judge on the held-out pairs

pair	category	gold	verifier	judge
V-H01	supported-verbatim	supported	✓	✓
V-H02	supported-verbatim	supported	✗	✓
V-H03	supported-verbatim	supported	✓	✓
V-H04	supported-verbatim	supported	✓	✓
V-H05	supported-verbatim	supported	✗	✓
V-H06	supported-verbatim	supported	✓	✓
V-H07	supported-verbatim	supported	✓	✓
V-H08	supported-verbatim	supported	✓	✓
V-H09	supported-paraphrase	supported	✓	✓
V-H10	supported-paraphrase	supported	✓	✓
V-H11	supported-paraphrase	supported	✓	✓
V-H12	supported-paraphrase	supported	✓	✓
V-H13	supported-paraphrase	supported	✓	✓
V-H14	supported-paraphrase	supported	✓	✓
V-H15	supported-paraphrase	supported	✓	✓
V-H16	relevant-not-supporting	unsupported	✗	✓
V-H17	relevant-not-supporting	unsupported	✓	✓
V-H18	relevant-not-supporting	unsupported	✗	✗
V-H19	relevant-not-supporting	unsupported	✗	✓
V-H20	relevant-not-supporting	unsupported	✓	✓
V-H21	wrong-number-or-id	unsupported	✓	✓
V-H22	wrong-number-or-id	unsupported	✓	✓
V-H23	wrong-number-or-id	unsupported	✗	✓
V-H24	wrong-number-or-id	unsupported	✓	✓
V-H25	wrong-number-or-id	unsupported	✗	✗
V-H26	negation-or-condition-flip	unsupported	✗	✓
V-H27	negation-or-condition-flip	unsupported	✗	✗
V-H28	negation-or-condition-flip	unsupported	✗	✗
V-H29	negation-or-condition-flip	unsupported	✓	✓
V-H30	negation-or-condition-flip	unsupported	✓	✓
V-H31	wrong-authority-or-scope	unsupported	✓	✗
V-H32	wrong-authority-or-scope	unsupported	✓	✓
V-H33	wrong-authority-or-scope	unsupported	✓	✓
V-H34	wrong-authority-or-scope	unsupported	✓	✓
V-H35	wrong-authority-or-scope	unsupported	✓	✓
V-H36	fabricated-quote	unsupported	✓	✗
V-H37	fabricated-quote	unsupported	✓	✗
V-H38	fabricated-quote	unsupported	✓	✓
V-H39	multi-citation	supported	✓	✓
V-H40	multi-citation	supported	✓	✓

E · Ablations, evidence level: labelled-invalid units in context per case

case	gov-minus-acl-scope	gov-minus-assembler	gov-minus-conflict	gov-minus-hybrid	gov-minus-lifecycle-authority	gov-minus-structured	gov-minus-verify	gov-erned	naive	naive-plus-acl-scope	naive-plus-assembler	naive-plus-hybrid	naive-plus-lifecycle-authority	naive-plus-structured	naive-plus-verify	nc-minus-recheck
H-K0	0	0	0	0	0	0	0	0	1	0	1	1	0	1	1	0
H-K1	1	0	0	0	0	0	0	0	4	0	3	4	2	4	4	0
H-K2	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
H-K3	0	0	0	0	1	0	0	0	3	1	3	3	0	3	3	0
H-K4	0	0	0	0	1	0	0	0	4	3	3	3	0	4	4	0
H-K5	0	0	0	0	0	0	0	0	5	4	4	4	0	5	5	0
H-K6	0	0	0	0	1	0	0	0	4	3	4	3	0	4	4	0
H-K7	1	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
H-K8	2	0	0	0	0	0	0	0	5	0	4	4	2	5	5	0
H-K9	0	0	0	0	0	0	0	0	1	1	1	2	0	1	1	0
H-K10	0	0	0	0	1	0	0	0	4	4	3	3	0	4	4	0
H-K11	0	0	0	0	0	0	0	0	2	0	2	2	2	2	2	0
H-K12	2	0	0	0	1	0	0	0	3	2	3	3	3	3	3	2
H-K13	0	0	0	0	0	0	0	0	4	3	4	5	0	4	4	0
H-K14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H-K15	0	0	0	0	0	0	0	0	1	1	1	1	0	1	1	0
H-K12b	1	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1
H-K12c	3	0	0	0	0	0	0	0	6	2	5	5	5	6	6	2
H-K14b	2	0	0	0	0	0	0	0	2	0	2	2	1	2	2	0
H-P1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H-P2	0	0	0	0	0	0	0	0	5	0	4	5	0	5	5	0
H-P3	0	0	0	0	0	0	0	0	1	0	1	1	0	1	1	0

PRODUCTION AI ENGINEERING · THE SERIES SO FAR

FOUNDATION

F1 · MCP Tool Sprawl

Your AI agent has 500 MCP tools. Now what?

FOUNDATION

F2 · Layered Architecture

Your agent works in a demo. Why does it break in production?

FOUNDATION

F3 · Headless AI

Your AI shouldn't live inside the UI

STATE

S1 · Memory, Context & State

Your agent remembers everything. That's a problem.

TRUST

T2 · Authorization & Policy

What is this agent actually allowed to do?

TRUST

T3 · Human-in-the-Loop

Authorized. Should it still act?

TRUST

T5 · Observability & Governance

Your AI agent did something in production. Can you explain exactly what happened?

TRUST

T4 · AI Control Plane

Your agents shouldn't govern themselves

TRUST

T1 · Agent Identity

Who is acting, and on whose authority?

TRUST

T6 · Agent, Tool & MCP Security

Assume the context is hostile.

RELIABILITY

R1 + R2 · Evals, Observability & Reliability

"The agent failed" is not an operational signal.

CAPSTONE

P1 · The Agent Is Not the Architecture

The final reference architecture

COORDINATION

C1 · Multi-Agent & A2A

Do you actually need multiple agents?

OPERATIONS

O1 + O2 · Operating AI Agents at Scale

Cost, latency, scale and lifecycle

CURRENT

S2 · Enterprise Knowledge & RAG

Your agent found the right document. Why did it give the wrong answer?

The full learning map (Start Here)

S2 · Production AI Engineering

Production AI Engineering · S2 · Run report. Every measured number here is substituted at build time from enterprise_knowledge_rag_poc/runs/2026-10-08-heldout/facts.json. Figures carry their provenance.

Medium edition Technical deep dive Evidence Check POC README QA report